SULIT

UTM UNIVERSITI TEKNOLOGI MALAYSIA
Fakulti Sains Komputer & Sistem Maklumat
UNIVERSITI PENYELIDIKAN

FINAL EXAMINATION SEMESTER I 2012/2013

SUBJECT CODE : **SCR1013 / SCSR1013**

SUBJECT TITLE : **DIGITAL LOGIC**

COURSE : **SCSR / SCSV / SCSI / SCSJ / SCSD**

DURATION : **3 HOURS**

DATE :

PLACE :

INSTRUCTIONS TO CANDIDATES:

1) **Please FILL in your particulars in the space provided below.**
2) **Answer ALL questions in space provided in the question paper.**
3) **This questions paper consists of Part A and Part B. Part A will contribute 35 marks while Part B contributes 65 marks.**

Name	
Matric No.	
Course	* SCSR / SCSV / SCSI / SCSJ / SCSD
Section	* 01 02 03 04 05 06 07 08
Lecturer's Name	

THIS PAPER CONTAINS 23 PAGES NOT INCLUDING THIS COVER PAGE.

5. The addition of an inverter to an SR flip flop creates _____.

A) JK flip flop

B) D flip flop

C) T flip flop

D) Master slave flip flop

6. A flip flop that can be built from JK flip flop by connecting J and K to the same node is _____.

A) SR flip flop

B) JK flip flop

C) T flip flop

D) D flip flop

7. The difference between SR flip flop and JK flip flop is SR has _____while JK doesn't.

A) Hold state

B) Toggle state

C) Invalid state

D) Clock

8. T flip flop has only 2 states, which are _____.

A) Set and Reset

B) Hold and Toggle

C) Set and Toggle

D) Reset and Toggle

14. Figure 4 shows timing diagram for an asynchronous counter using J-K flip flop. Choose the **FALSE** statement describing the counter with Q_0 initially LOW.

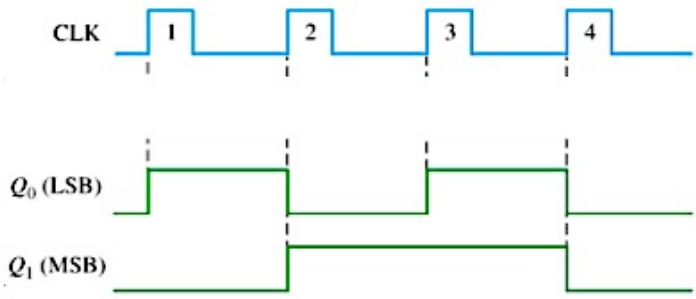


Figure 4

A) The counting sequence is unidirection up.

B) The type of J-K flip flops used is positive edge.

C) Q_1 is complement when Q_0 = HIGH after next clock edge.

D) It is a MOD-2 counter.

16. Two cascaded decade counters is shown in Figure 6. Choose the **FALSE** statement about the counter.

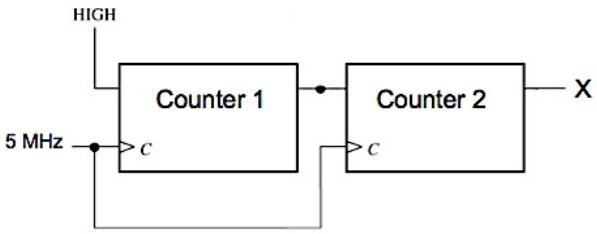


Figure 6

A) The counter performs as frequency divider.

B) Total flip flop used is 4.

C) It also knows as modulus-100 cascaded counter.

D) To obtain X = 0.005 MHz, the circuit needs one more similar counter to be cascaded.

19. A MOD-10 Johnson counter requires _____.

A) 10 flip-flops

B) 4 flip-flops

C) 5 flip-flops

D) 12 flip-flops

23. Choose the **TRUE** statement of Gated SR latch.

i. It's output change at the edge of the clock

ii. ENABLE pin has a higher priority than pin S and R.

iii. It has CLK to control its operation.

iv. If EN = 1, the output will change immediately according to input S and R.

A) ii and iv.

B) i, ii and iii.

C) iii only.

D) iv only.

24. Choose the **TRUE** characteristics for asynchronous input of a flip flop.

i. Clock has the highest priority than asynchronous input $\overline{PRE}$ and $\overline{CLR}$.

ii. If $\overline{CLR}$ = 0, reset the output to '0'.

iii. If $\overline{PRE}$ = LOW, reset the output to LOW.

iv. Asynchronous input $\overline{PRE}$ and $\overline{CLR}$have the highest priority than Clock.

A) i and ii.

B) ii and iv.

C) i and iii.

D) iii and iv.

25. Circuit in Figure 7 is using a Full Adder (FA). Choose the **TRUE** statement about the circuit.

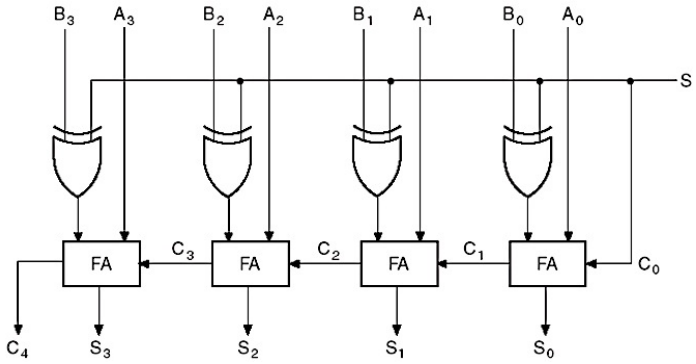


Figure 7

i. It can add two 4 bits numbers.

ii. If S HIGH, B will be subtracted from A.

iii. The arithmetic uses a 1's complement.

iv. XOR always function as an inverter in the circuit.

A) i and ii.

B) i,ii and iii.

C) iii and iv.

D) i,ii, iii and iv.

27. Figure 8 illustrates a decoder that connected to an asynchronous counter using D flip flop. Choose the **TRUE** statements about the counter.

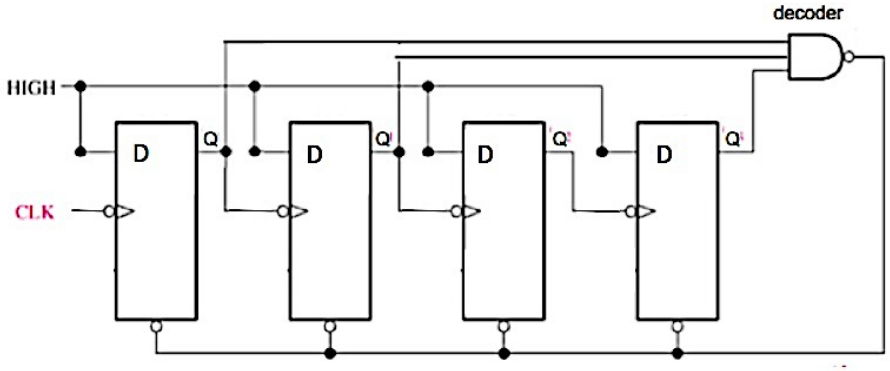


Figure 8

i. It is a count down counter.

ii. The decoder will reset the counter when its input is 1011_2.

iii. The $\overline{CLR}$ input will reset when output of the decoder is 1.

iv. The counter will have eleven states.

A) i, iii and iv.

B) ii and iv.

C) i and iii.

D) i, ii, iii, and iv.

CONFIDENTIAL

FINAL EXAMINATION SEMESTER I 2013/2014

SUBJECT CODE : SCSR1013

SUBJECT TITLE : DIGITAL LOGIC

YEAR/COURSE :

DURATION :

DATE :

VENUE :

QUESTION 2 (20 Marks)

(a) Referring to Figure 4, answer the questions below.

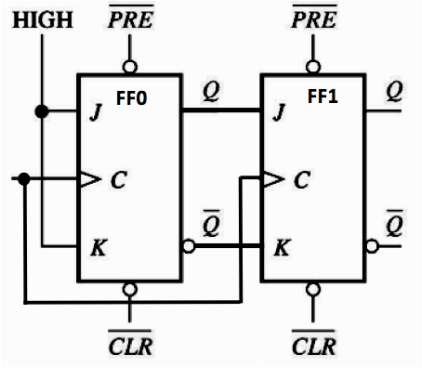


Figure 4

i. What are the possible values of the signals to activate the asynchronous operation? [2 marks]

ii. JK flip-flop is a universal flip-flop, which can be configured to work as different flip-flops. What type of flip-flop do FF1 and FF0 represent? Write your answers and provide justification in Table 4. [5 marks]

Table 4

Flip-flop	Type	Justification
FF1		
FF0		

iii. Fill in Table 5 with the appropriate inputs according to the flip-flop priority operation. [2 marks]

Table 5

Priority	Input(s)
Highest	
Lowest	

iv. Complete the values in Table 6 after 3 clock cycles for the given conditions. The initial value of Q_{FF0} is 0. Justify your answer. [5 marks]

Table 6

FF0		FF1		CLOCK	Q_{FF0}	Q_{FF1}
$\overline{PRE}$	$\overline{CLR}$	$\overline{PRE}$	$\overline{CLR}$			
1	0	1	1			

Justification:

(b) Draw the output, Q of the different latches and flip-flop in Figure 5. Assume positive-edge clock. [6 marks]

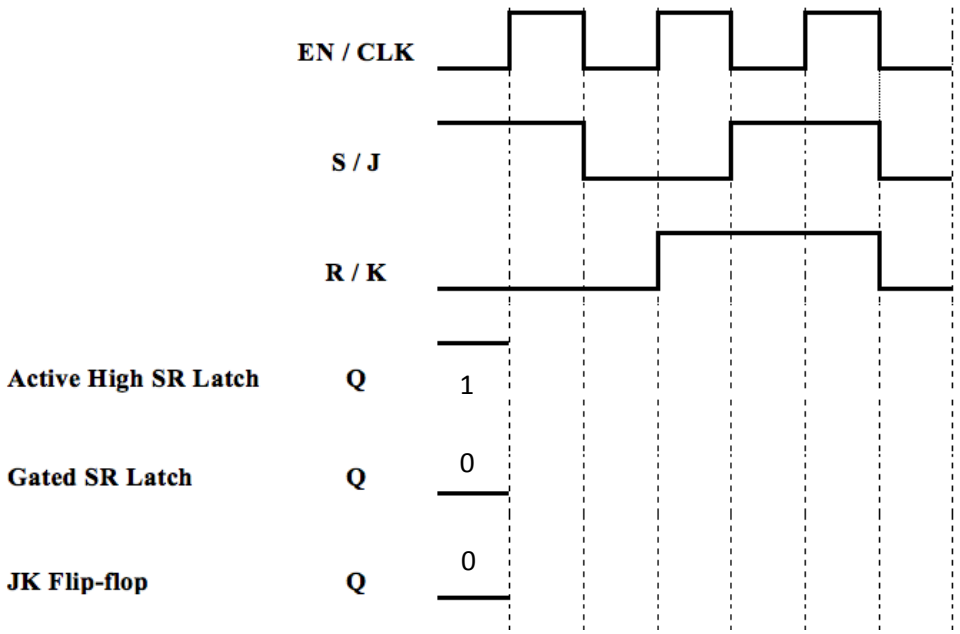


Figure 5

QUESTION 3 (20 Marks)

Answer the following questions based on Figure 6.

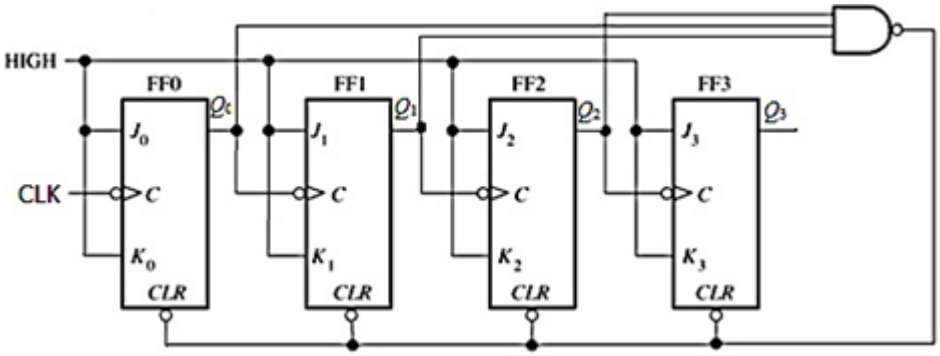


Figure 6

(a) Draw the complete timing diagram for ten clock pulses in Figure 7. Initial value for $Q_3 = Q_2 = Q_1 = Q_0 = 0$. [4 marks]

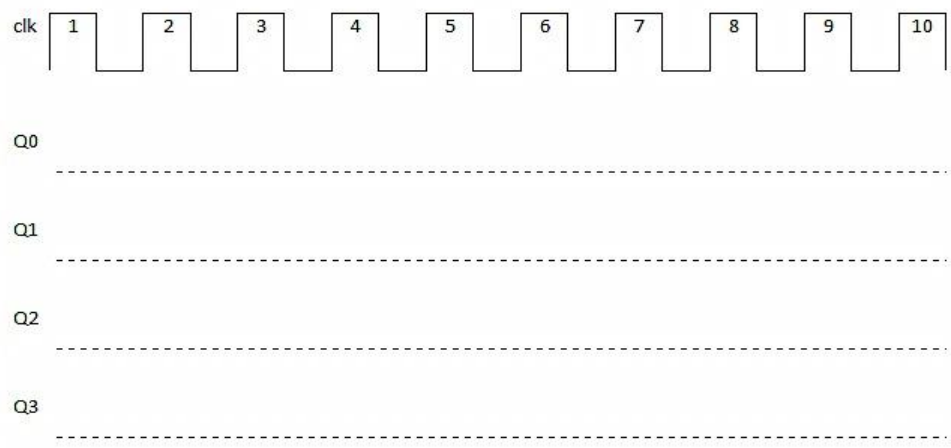


Figure 7

(b) Determine whether the counter is asynchronous or synchronous. Justify. [3 marks]

Type of counter:

Justification:

(c) Is the counter count-up or count-down? [1 mark]

(d) What is the minimum and maximum number that can be counted by the counter based on **ONLY** four flip-flops? [2 marks]

(e) What is the purpose of the NAND gate in Figure 6? [2 marks]

(f) What is the actual modulus (MOD) for the counter? [2 marks]

(g) From your answer in 3(f), draw the state diagram for the counter. [4 marks]

(h) Based on your answers, comment on the efficiency of the counter design. [2 marks]

SUBJECT CODE	:	SCSR1013
SUBJECT NAME	:	DIGITAL LOGIC
DURATION	:	**2 HOURS 45 MINUTES**

(Instructions to Candidates) :

1. Please answer PART A in the OBJECTIVE ANSWER SHEET given in PAGE 6 of this question booklet.
2. Please answer Questions 2 (a, b, c) and 5 (c) of PART B in the TIMING DIAGRAMS in this question booklet.
3. Please answer ALL OTHER questions of PART B in the ANSWER BOOKLET given.

THIS QUESTION PAPER CONSISTS OF THIRTEEN (13) PRINTED PAGES NOT INCLUDING THIS FRONT PAGE

5. MUX is also known as _____.

A. data distributor

B. data selector

C. adder

D. decoder

6. Which of the following is NOT a latch?

A. S-R latch

B. J-K latch

C. Gated S-R latch

D. Gated D latch

7. The invalid state for S-R latch occurs when S and R are _____.

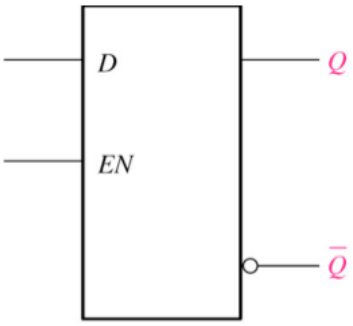
A. both high

B. both low

C. S = high, R = low

D. S = low, R = high

8. The figure below shows what type of temporary storage device?



A. D flip flop

B. Gated-D latch

C. Gated S-R latch

D. C-D flip flop

9. Which flip-flop is versatile and widely used?

A. Edge-triggered D

B. S-R

C. J-K

D. T

10. Choose the CORRECT statement.

A. Synchronous counter has clocks from the same source.

B. Synchronous counter has clocks from the different sources.

C. Asynchronous counter has clocks from the same source.

D. Asynchronous counter doesn’t have clock input.

14. If a propagation delay of an asynchronous recycle UP counter using FIVE (5) J-K flip-flop is 2 seconds, what is the maximum operating frequency?

A. 0.1 Hz

B. 0.2 Hz

C. 0.5 Hz

D. 2 Hz

15. The table below shows an excitation table for which flip-flop?

Q_{n+1}	Input flip-flop
Q_n	0
$\overline{Q_n}$	1

A. D

B. T

C. J-K

D. None of the above

19. A MOD 8 ring counter requires how many flip-flops?

A. 2

B. 4

C. 8

D. 16

QUESTION 2 [15 MARKS]

*** Answer Questions (a), (b) & (c) in the Timing Diagrams of Figures 2 to 4 in this Question Booklet.**

a) Determine the timing diagram in Figure 2 for output Q for $\overline{S} - \overline{R}$ latch. Assume that Q is initially **LOW**. Draw your answer in Figure 2. [3m]

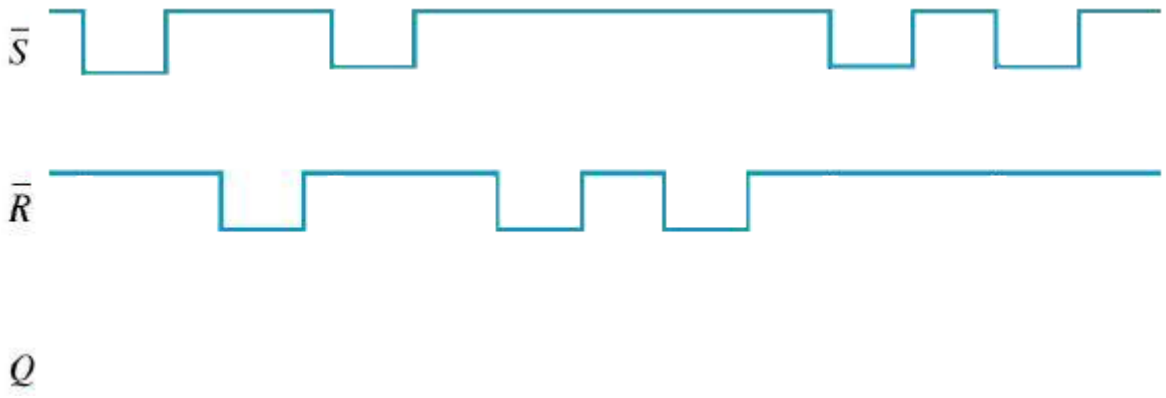


Figure 2: Timing Diagram for $\overline{S} - \overline{R}$ Latch

b) Determine the timing diagram in Figure 3 for output Q for gated S-R latch. Assume that Q is initially **LOW**. Draw your answer in Figure 3. [3m]

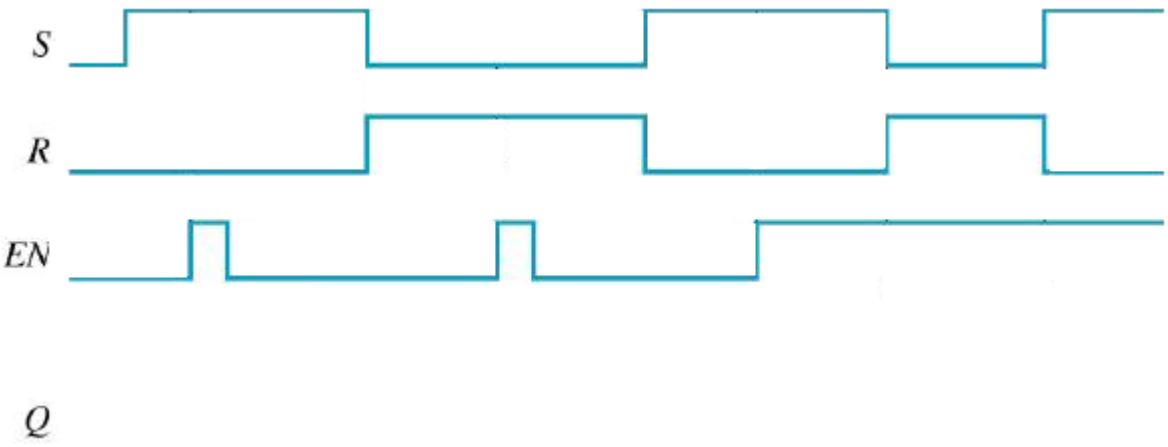
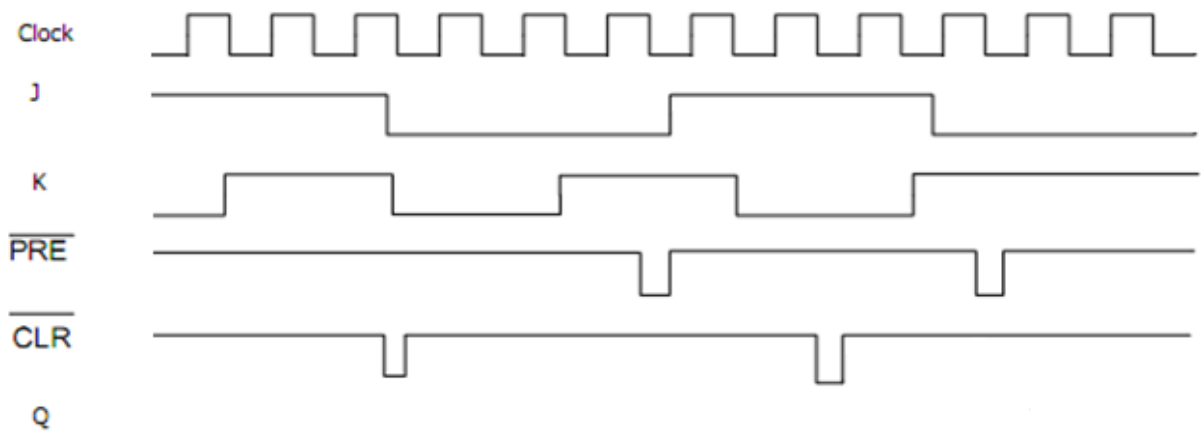


Figure 3: Timing Diagram for Gated S-R Latch

c) Determine output Q in Figure 4 for a positive edge triggered clock J-K flip-flop with $\overline{PRE}$, $\overline{CLR}$, and J, K inputs. Assume that Q is initially **LOW**. Draw your answer in Figure 4. [5m]



QUESTION 3 [10 MARKS]

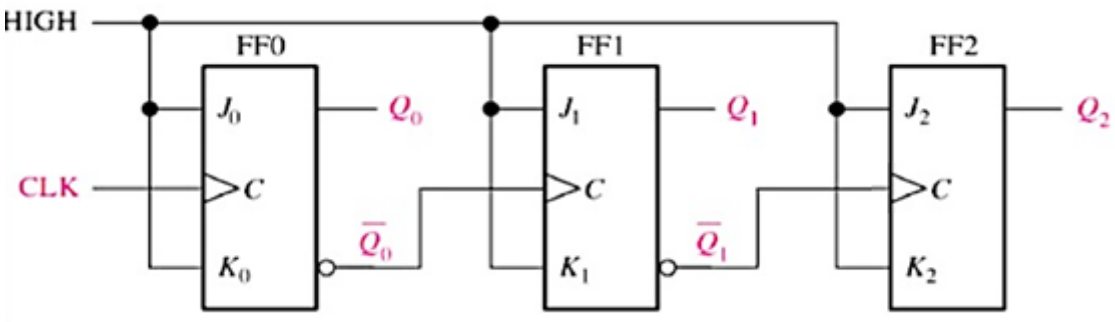


Figure 5: Connection of Logic Symbol for an Asynchronous Binary Counter

a) Based on Figure 5, draw the state diagram for the counter by using binary representation. [4m]

b) Redesign the connection of logic symbol for a:

- 3-bit count UP ripple counter,
- asynchronously clocked (truncated) MODULUS-5
- using J-K flip-flop
- with NEGATIVE edge triggered clock. [6m]

~ END OF QUESTION 3 ~

SCSR 1013 DIGITAL LOGIC FINAL EXAM 2015/2016-1

OBJECTIVE QUESTIONS (30 POINTS)

Answer the questions by filling your answers on PAGE 18.

1. Which of the following is **NOT** a latch?

   A. S-R latch
   B. J-K latch
   C. Gated S-R latch
   D. Gated D latch

2. The invalid state for S-R latch occurs when S and R are _____.

   A. both high
   B. both low
   C. S = high, R = low
   D. S = low, R = high

3. With regard to a D latch, _____.

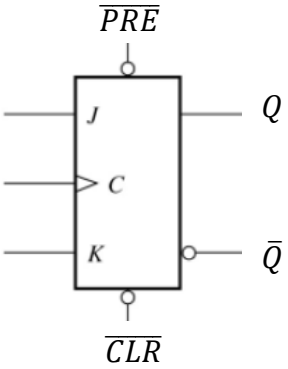
   A. the Q output follows the D input when EN is LOW
   B. the Q output is opposite the D input when EN is LOW
   C. the Q output follows the D input when EN is HIGH
   D. the Q output is HIGH regardless of EN's input state

4. When **both inputs** of a J-K pulse-triggered flip-flop are **LOW**, and the clock triggers, what will the output be?

   A. An invalid state will exist.
   B. No change will occur in the output.
   C. The output will toggle.
   D. The output will reset.

SCSR 1013 DIGITAL LOGIC FINAL EXAM 2015/2016-1

5. Refer to the following flip-flop, which of the statement is **FALSE**?



A. J and K are two input of the flip-flop.

B. $\overline{PRE}$ and $\overline{CLR}$ are the highest input priority.

C. In the synchronous mode, both $\overline{PRE}$ and $\overline{CLR}$ must be set to HIGH.

D. In the asynchronous mode, both $\overline{PRE}$ and $\overline{CLR}$ must be set to LOW.

6. Given an asynchronous counter with MOD 13, which of the statements is **TRUE** regarding the counter?

A. The counter is a truncated counter.

B. The counter is a 3-bit counter.

C. The counter consists of gated D latch.

D. Clock signal of all flip-flops are coming from the same source.

8. What is the clock maximum frequency limit of an asynchronous counter, which consists of 4 Flip-Flop, with each FF propagation delay is equal to 4ms?

   A. 62.5 Hz
   B. 16 Hz
   C. 0.25 Hz
   D. 25 Hz

9. The last state of a modulus-11 binary counter is _____.

   A. 1011
   B. 1010
   C. 1001
   D. 1100

10. The designation $UP/\overline{DOWN}$ means that the _____.

   A. up count is active-HIGH, the down count is active-LOW
   B. up count is active-LOW, the down count is active-HIGH
   C. up and down counts are both active-LOW
   D. up and down counts are both active-HIGH

11. The sentence "the data only *appear at the output in parallel*, at the positive edge of the clock" refers to

   A. Serial In Serial Out (SISO) shift register
   B. Single In Bit Out (SIBO) shift register
   C. Parallel In Serial Out (PISO) shift register
   D. Parallel In Parallel Out (PIPO) shift register

SCSR 1013 DIGITAL LOGIC FINAL EXAM 2015/2016-1

STRUCTURE QUESTIONS (70 POINTS)

QUESTION 1 (13 POINTS)

A. Determine the timing diagram in Figure 1 for output Q for gated S-R latch. Assume that Q is initially **LOW**. Draw your answer in Figure 1. [3 pts]

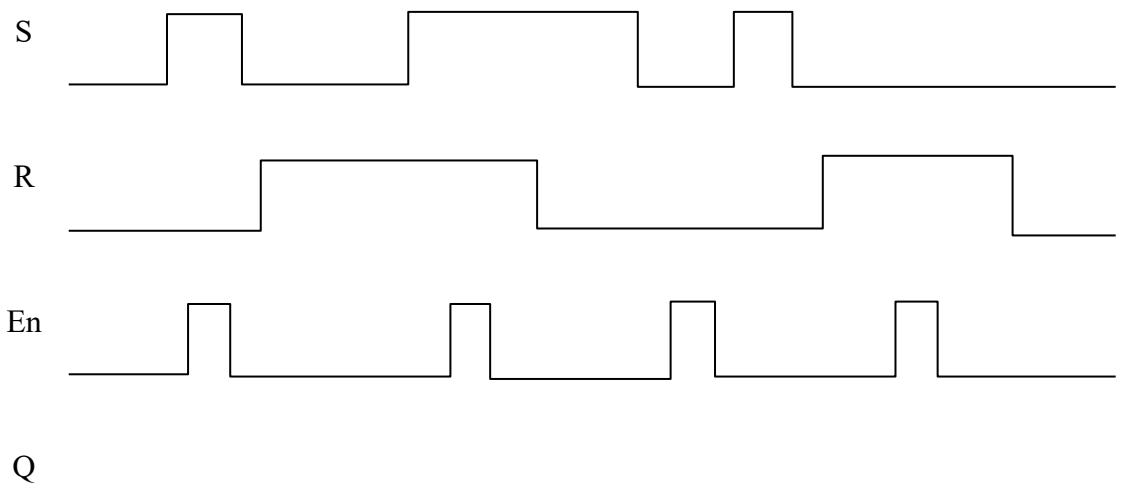


Figure 1. Timing Diagram for Gated S-R Latch

B. Determine the timing diagram in Figure 2 for output Q for $\overline{S} - \overline{R}$ latch. Assume that Q is initially **LOW**. Draw your answer in Figure 2. [3 pts]

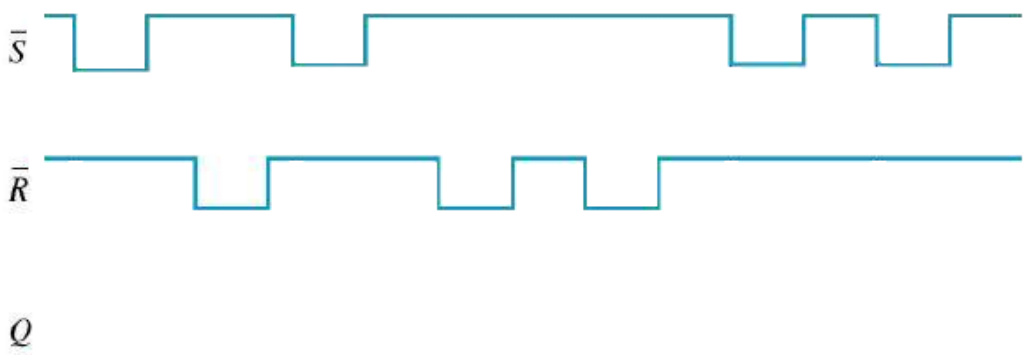


Figure 2. Timing Diagram for $\overline{S} - \overline{R}$ Latch

C. Refer to Figure 3. Given a positive edge triggered JK flip-flops, complete the following timing diagram. [3 pts]

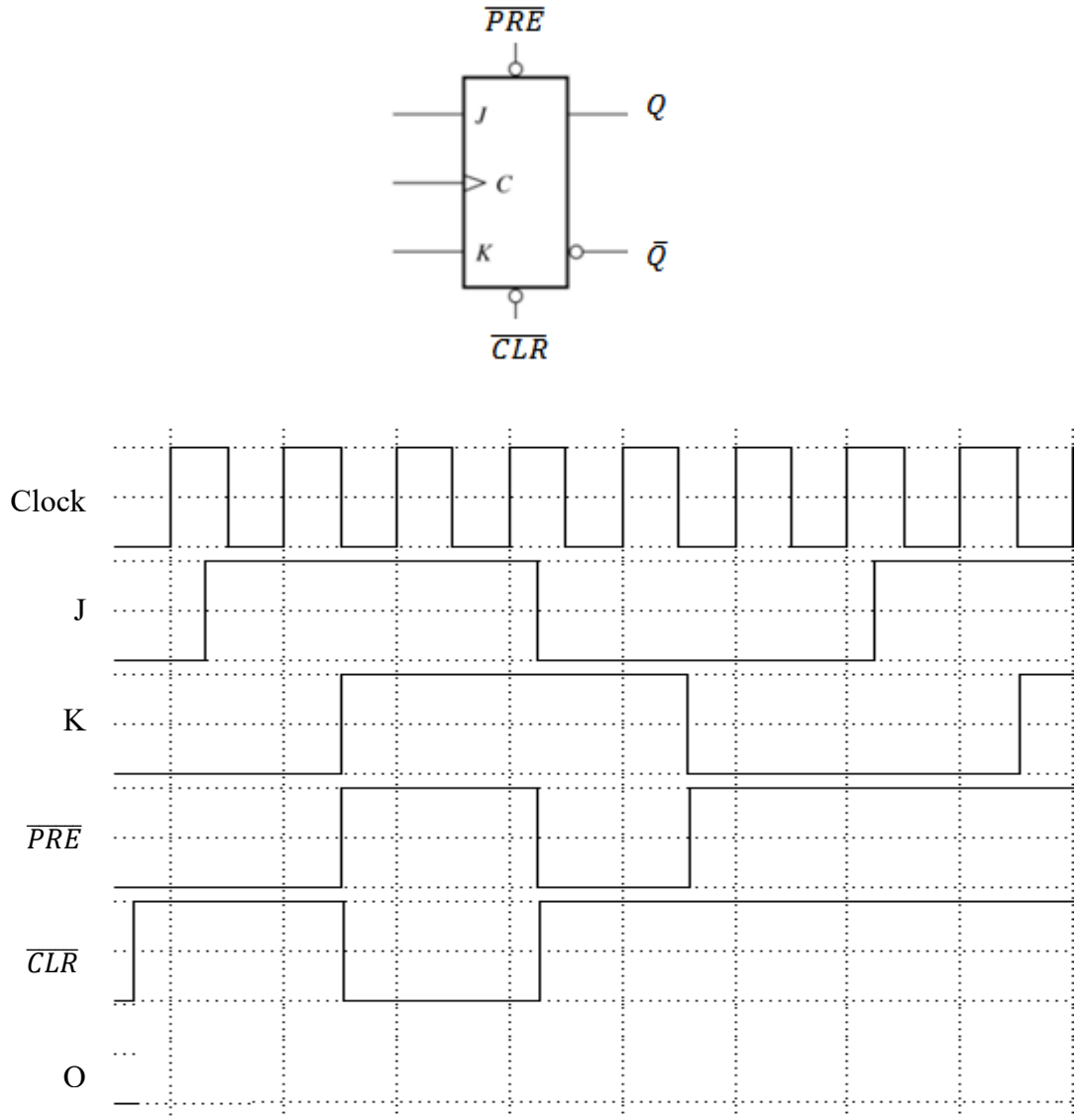


Figure 3

SCSR 1013 DIGITAL LOGIC FINAL EXAM 2015/2016-1

QUESTION 2 (15 POINTS)

A. Design an asynchronous counter that has the following properties: [6 pts]

i. The counter is a MOD 6 counter.

ii. The counter sequence direction is counting up.

iii. The counter consists of active-low JK FF.

SCSR 1013 DIGITAL LOGIC FINAL EXAM 2015/2016-1

QUESTION 3 (10 POINTS)

A. Draw 3-bit Ring Counter and Johnson Counter using D flip flop to show the significant difference between them. [6 pts]

B. How many state produced for the 3-bit Johnson Counter and Ring Counter? [2 pts]

SULIT

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION SEMESTER I, 2016 / 2017

SUBJECT CODE :

SUBJECT NAME :

SECTION :

TIME :

DATE/DAY :

VENUES :

INSTRUCTIONS :

Answer all questions from **Part A** and **B** in this question booklet. For **Part B**, read the questions carefully and show **ALL** your works in details.

This examination will contribute 35% towards the total marks of 100 points.

Warning!
Students who are caught cheating during the examination will be reported to disciplinary board for action to suspend the student for one or two semesters.

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of ____ (__) printed pages excluding this page.

6. Which of the following statements is **FALSE** about a latch and a flip-flop?

A) Latch and flip-flop can store binary bit 0 or 1.

B) A latch requires a clock input before output latch can be changed.

C) A flip-flop output will change according to the input flip-flop at the clock edge triggered.

D) A latch output will change according to the input level of the latch.

7. Which of the following statements is **FALSE** about an S-R latch?

A) It has a SET state.

B) It has a RESET state.

C) It has a HOLD state.

D) It has a TOGGLE state.

8. How many different states does a 3-bit asynchronous counter has?

A) 3

B) 6

C) 8

D) 9

10. What is the maximum state of a typical MOD 10 (decade) binary counter?

A) 0000
B) 1010
C) 1111
D) 1001

11.

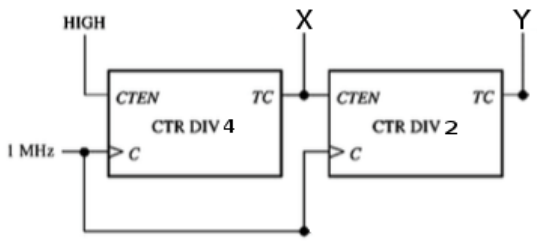


Figure 2

Referring to Figure 2, what is the signal frequencies of X and Y?

A) X = 0.25 MHz, Y = 0.125 MHz.
B) X = 4 MHz, Y = 2 MHz.
C) X = 0.125 MHz, Y = 0.25 MHz.
D) X = 2 MHz, Y = 4 MHz.

12. Which of the following statements is **TRUE** about synchronous counter?

A) Synchronous counter can count up and count down simultaneously (with the same clock cycle).
B) If synchronous counter is implemented with N flip-flops in its sequential circuit, the number of valid states of the counter is always 2^N.
C) Synchronous counter can be implemented using D flip-flops, as long as the clock source of the flip flops is taken from the common clock source.
D) Synchronous counter can be implemented using any type of flip-flop, as long as the clock source of flip flops is taken from the output of the preceding flip-flops.

13. The following are Shift Registers, **EXCEPT**:

A) SISO Serial In, Serial Out
B) SIPO Serial In, Parallel Out
C) SILO Serial In, Last Out
D) PIPO Parallel In, Parallel Out

14. The following statements are true about Shift Register (SR) **EXCEPT**:

A) SR stores binary data
B) SR performs data rotate
C) SR consists of several latches
D) SR is used for data transfer

15. A **MOD 10** Ring counter requires _____.

A) ten flip-flops
B) five flip-flops
C) four flip-flops
D) eight flip-flops

Question 2 [15 Marks]

(a) Figure 6 shows a basic circuit of a latch. Answer the following based on the circuit.

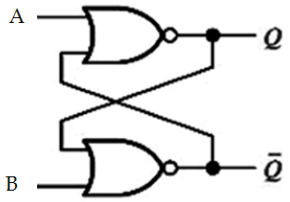


Figure 6

(i) Complete the following truth table with initial values of Q and $\overline{Q}$ are bit 1 and 0. [4M]

Input		Output		State
A	B	Q	$\overline{Q}$	
0	0			
0	1			
1	0			
1	1			

(ii) What type of latch is represented by the circuit in Figure 6? [1M]

(b) Referring to Figure 7 in page 12, complete the output $\boldsymbol{Q}$ for a Gated $\overline{S}$ - $\overline{R}$ latch. Label each state that occurs for each transition in the timing diagram. [3M]

(c) Referring to Figure 8, answer the following questions. Assume Q is initially low.

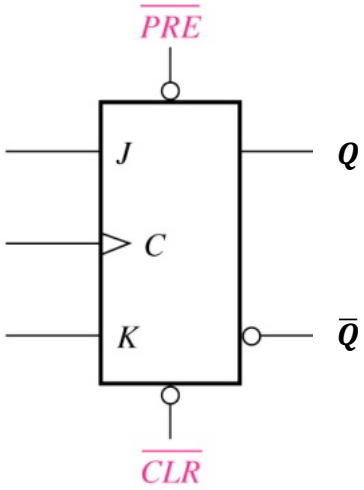


Figure 8

(i) Draw output $\overline{\boldsymbol{Q}}$ in the timing diagram at page 12. [3M]

(ii) List the input priority of the flip-flop from the highest to the lowest. [2M]

(d) Draw the implementation of negative edge T flip-flop by using JK flip-flop. [2M]

Question 3 [40 Marks]

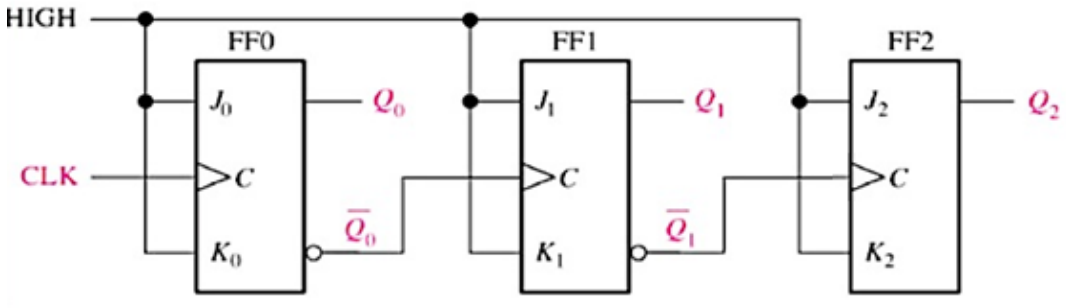


Figure 9: Asynchronous binary counter

(a) Based on Figure 9, draw the State Diagram for the counter by using binary representation. [4M]

(b) Redesign Figure 9, by considering new requirements as follows:

- 3-bit count up ripple counter with Modulus-5;
- using J-K Flip-Flop and
- with negative edge triggered clock. [6M]

(c) Rolek Corp is designing its new 2-bit binary counter. The state diagram of the counter is given in Figure 10:

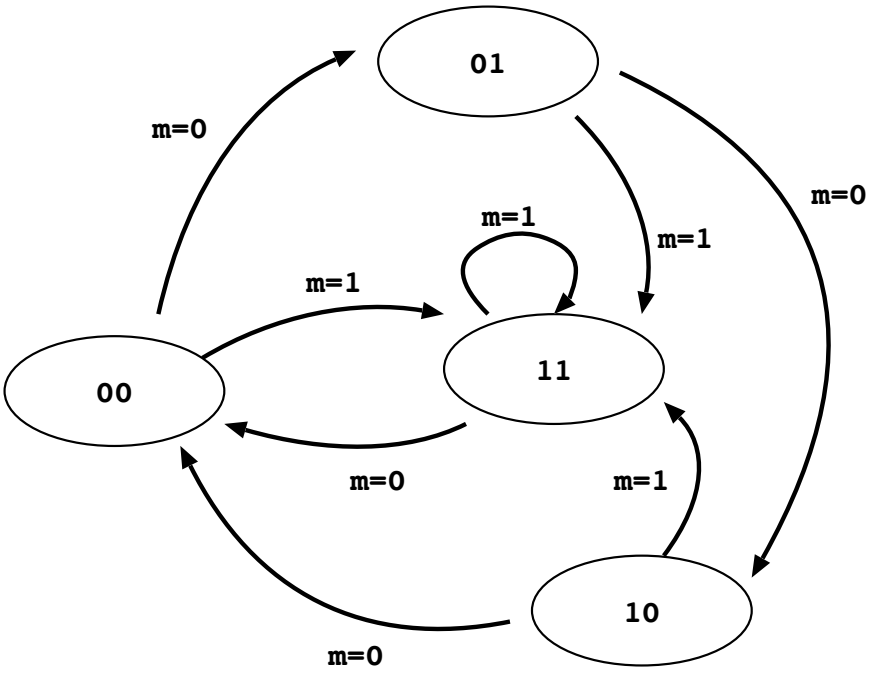


Figure 10: State Diagram for 2-bit Rolek binary synchronous counter

ATTACHMENT B

Answer for Question 1 (d)

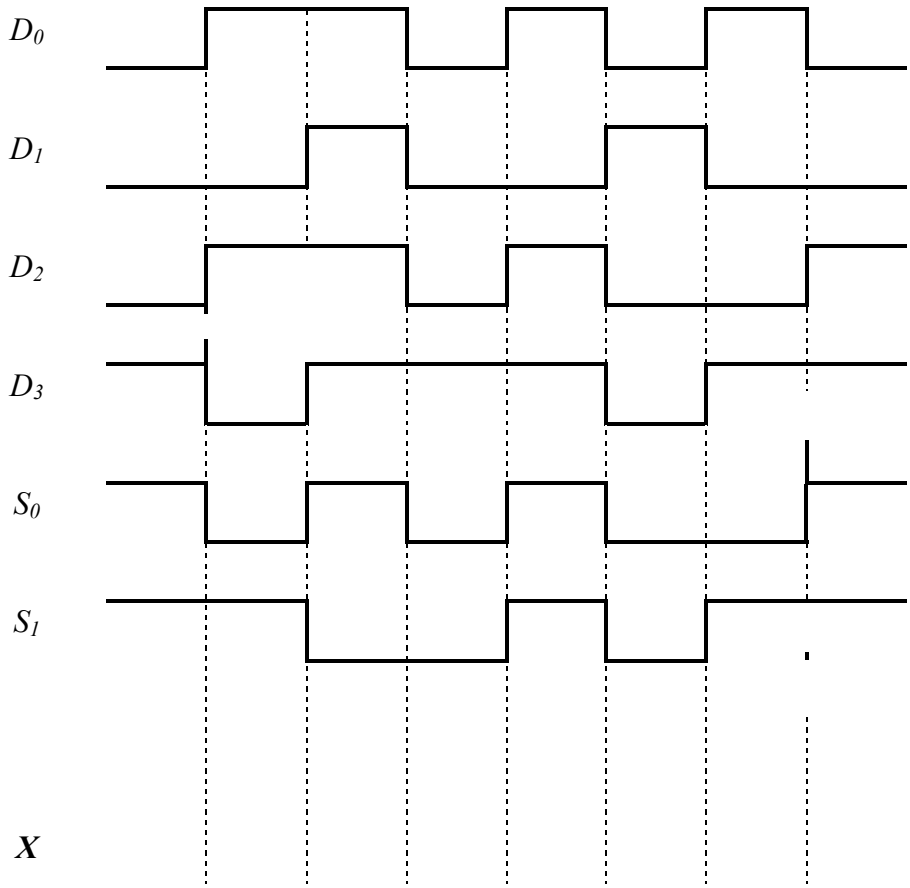


Figure 5

Answer for Question 2 (b)

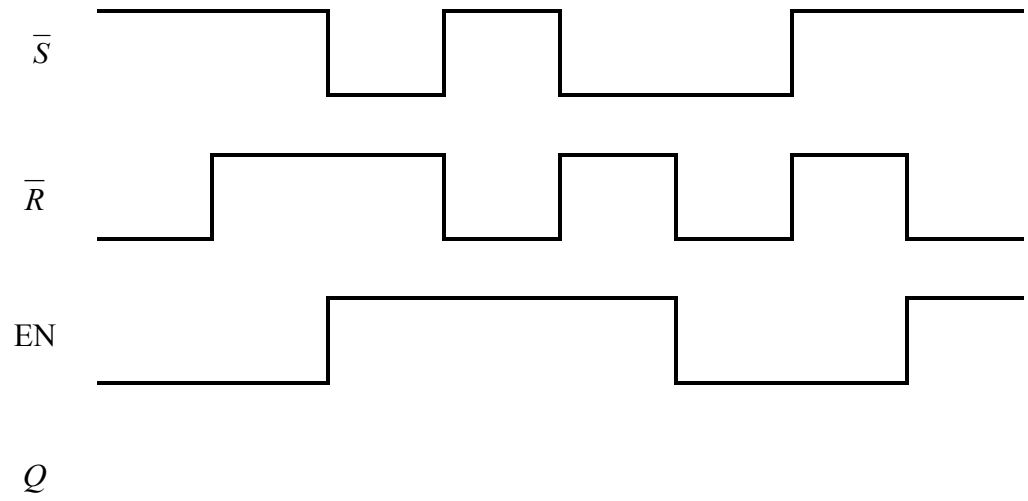


Figure 7

Answer Question 2 (c)(i)

